NAGENDRAKUMAR VENGISETTY

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PROFESSIONAL SUMMARY

AI/ML Developer with hands-on experience building production-grade AI systems — including real-time multi-model proctoring pipelines, LLM-based evaluation systems, NLP extraction pipelines, and recommendation engines — using computer vision, transformer embeddings, vector search, and speech/entity recognition. Skilled in end-to-end delivery across Python, FastAPI, Celery, and REST API development, with a track record of measurably improving accuracy, efficiency, and scalability across systems deployed to 100,000+ users.

PROFESSIONAL EXPERIENCE

AIML Developer | Altrodav Technologies Pvt Ltd (MeetMux & PlaceMux) / *Apr 2025 – Present* / *Bangalore, India*

PlaceMux — AI/ML Systems (Current Project)

Tech Stack: FastAPI, Celery, Redis, PostgreSQL, Python, InsightFace, YOLOv8, Silero VAD, Resemblyzer

- Built and hardened a real-time AI proctoring system for large-scale technical assessments, integrating multi-model inference (InsightFace for face recognition, YOLOv8 for object detection, Silero VAD and Resemblyzer for voice activity/speaker detection) across a FastAPI + Celery + Redis pipeline.
- Resolved critical production issues including SSE race conditions, silent YOLO crashes, missing Celery queue registrations, and migrated inference models from CPU to GPU for improved throughput.
- Engineered an AI coding evaluation system combining static analysis (Tree-sitter, Semgrep) with LLM-based evaluation (Qwen 2.5 7B), splitting Celery workers by workload type and setting scoring weights to prioritize runtime correctness (60/20/20), improving evaluation accuracy and system reliability to 93%.
- Delivered a JD Skill Extractor pipeline (GLiNER, Aho-Corasick, Qwen2.5:7b fallback) for automated job description parsing, improving extraction accuracy from 72% to 87% through ratio-based LLM triggering, an expanded blacklist (~278 phrases), and a 216-trigger implication graph.
- Developed a Domain Level Prediction system for technical interview questions, fixing model and dataset issues (524 duplicate entries, sparse difficulty bands) and generating synthetic data for underrepresented bands, achieving 94.2% domain accuracy and Cohen's Kappa of 0.94.

MeetMux — AI Developer (Social Platform)

Tech Stack: Python, FastAPI, Transformer Embeddings, Amazon Nova Sonic, BERT NER, Vector Search

- Designed an AI-powered chat reply recommendation system that generates context-aware responses, improving reply relevance and conversational accuracy by 35–40%.
- Implemented semantic similarity using Transformer-based embeddings and vector search to retrieve top replies from 100K+ chat pairs, achieving 90%+ semantic match accuracy.
- Architected a multi-layer recommendation pipeline combining personal history, friend-specific context, and global fallback, reducing cold-start failures by 30%.
- Enhanced response quality by applying toxicity filtering, spam detection, and similarity-based deduplication, reducing unsafe or repetitive replies by 25%.
- Created an AI-powered user onboarding assistant that enables users to complete their profiles through voice-based conversations, improving onboarding efficiency by 40%.
- Integrated Amazon Nova Sonic for high-speed intent matching and BERT NER for accurate entity extraction (e.g., name, DOB, bio) from user responses, achieving 95% entity accuracy.
- Automated the profile form-filling process by mapping extracted entities to application fields, eliminating 100% of manual entry steps and ensuring data consistency.
- Improved user experience and retention by reducing onboarding drop-offs by 25% through a seamless conversational interface.

PROJECTS

Full Stack e Commerce Application

- Developed a responsive e Commerce platform using React.js and Django, supporting 100+ products and 50+ active users in the initial phase.
- Architected and implemented RESTful APIs using Django REST Framework (DRF), enabling seamless data flow between the frontend and backend with latency reduced by 15% and improving API response times by 20%.
- Crafted a responsive mobile interface using React Bootstrap, achieving seamless compatibility for over 100 product listings and ensuring optimal user engagement across both desktop and mobile platforms throughout initial launch phase.
- Optimized application scalability and maintainability by adhering to best practices, improving performance by 15%.

Employee Management System (React.js + Spring Boot)

- Built a full-stack employee management system using React.js and Spring Boot for managing 50+ employee records.
- Added CRUD features to create, view, update, and delete records with real-time updates, reducing delay by 40%.
- Connected frontend and backend using REST APIs, improving data transfer speed by 30%.
- Designed a responsive interface with form checks and live table updates, reducing user input errors by 25%.

IMDB Clone: Movie Review and Streaming Platform

- Created an IMDB-inspired platform with Django, supporting a movie database with 100+ titles and user-generated reviews.
- Strengthened security with token-based authentication, reducing unauthorized access by 80%.
- Optimized performance by implementing throttling, pagination, and filtering, increasing page load speed by 50%.
- Restricted review editing permissions to review owners, ensuring 100% data integrity and user privacy.

Deep Learning Based Breast Cancer Classification System

- As a fourth-year project, participated in a team of four to develop an innovative system for breast cancer classification.
- Incorporated transfer learning with a pre-trained DenseNet121 model for feature extraction.
- Trained a model to categorize breast cancer as benign, malignant, or normal, achieving an exceptional 98% accuracy.

EDUCATION

Godavari Institute of Engineering and Technology, Rajahmundry | *2020 – 2024*
B.Tech, CSE (AIML) — CGPA: 8.42

Sri Vagdevi Junior College, Kodad | *2018 – 2020*
Intermediate, MPC — 91%

SKILLS & CERTIFICATIONS

Technical Skills

Programming Languages: Python, Java

AI/ML & Data Science: Machine Learning, Deep Learning, Neural Networks, Natural Language Processing (NLP)

Generative AI: LLMs, RAG, Prompt Engineering, LangChain, LangGraph, MCP, Vector Search, vLLM, Ollama

AI/ML Frameworks & Tools: LLM Evaluation

Backend & APIs: FastAPI, Django, Django REST Framework, Spring Boot, REST API Design

Task Queues & Async Processing: Celery, Redis

Databases: PostgreSQL, MySQL

Tools: Docker, Git

Frontend: React.js, HTML, CSS

Certifications: Python (Microsoft), Java Basic (HackerRank), Coding (Accenture)